

3 **(long)** The Humphrey cycle, which models pulse detonation engines, consists of the following four thermodynamic processes: (i) a gas is compressed isentropically from states 1 to 2, (ii) heat is added at constant volume from states 2 to 3, (iii) the gas is expanded isentropically from states 3 to 4, (iv) heat is rejected at constant pressure from states 4 to 1. This is not a steady flow process. You may consider the gas to be perfect with fixed c_p and $\gamma = 7/5$ throughout the cycle. T_i , p_i and v_i are the temperature, pressure and specific volume, respectively, at each state i .

(a) The pressure ratio, p_2/p_1 , of process (i) is denoted r . Write down expressions for T_2/T_1 and v_2/v_1 in terms of r and sketch the cycle on a $T - s$ diagram. [4]

(b) Use a TdS equation to show that $d(\ln T)/ds = 1/c_v$ during process (ii). Express $T_3 - T_2$ in terms of the specific heat added, q , and the specific heat at constant volume, c_v . Show that the change in specific entropy is $s_3 - s_2 = c_v \ln(1 + q/(c_v T_2))$. [6]

(c) Use a TdS equation to show that $d(\ln T)/ds = 1/c_p$ during process (iv). [2]

(d) With reference to your previous answers, carefully re-sketch this Humphrey cycle on a new $T - s$ diagram. [6]

(e) The Brayton cycle is identical to the Humphrey cycle but with heat added at constant pressure. For the same pressure ratio, r , and specific heat added, q , calculate T_3 and $s_3 - s_2$. By considering $d(s_3 - s_2)/dq$, or otherwise, show that $s_3 - s_2$ is always larger for the Brayton cycle than for the Humphrey cycle and carefully sketch the Brayton cycle on the same $T - s$ diagram as the Humphrey cycle of part (d). [6]

(f) By inspecting the $T - s$ diagram, and without further calculations, state which cycle is more thermodynamically efficient. Explain your answer. [6]